

Nmap TestMe Vulnerability Assessment Techniques

Assessment of scanme.nmap.org from Kali Linux

1. Assessment Overview

A network vulnerability assessment was performed from a Kali Linux virtual machine against scanme.nmap.org, the Nmap-provided test target. The assessment used Nmap to demonstrate host discovery, service and version enumeration, and vulnerability-oriented NSE scripting. The scan identified the target as 45.33.32.156 and confirmed that the host was reachable.

The purpose of this assessment was to demonstrate vulnerability-assessment techniques and document the security exposure identified through Nmap. The results below are based on the terminal evidence and screenshots collected during the assessment.

2. Assessment Methodology

Technique	Command	Purpose
Host Discovery	<code>nmap -sn scanme.nmap.org</code>	Determines whether the authorized target is reachable and identifies its resolved IP address.
Service and Version Enumeration	<code>nmap -sV scanme.nmap.org</code>	Identifies open/filtered ports and attempts to determine the service and software version operating on each port.
Vulnerability Assessment	<code>nmap -sV --script vuln scanme.nmap.org</code>	Uses Nmap's vulnerability-oriented NSE scripts to identify potential weaknesses, version/CPE vulnerability matches, and web-security observations.
Result Preservation	<code>nmap -sV scanme.nmap.org -oN scanme-services.txt</code>	Saves the service-enumeration results to a text file for documentation and later analysis.

3. Asset Discovery Results

The host-discovery scan identified scanme.nmap.org as an active host. Nmap resolved the target to IPv4 address 45.33.32.156 and reported the host as up with approximately 0.073 seconds of latency. An IPv6 address was also displayed by Nmap but was not scanned.

```
(root@kali)~# nmap -sn scanme.nmap.org
Starting Nmap 7.94 ( https://nmap.org ) at 2026-09-09 12:55 PDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.073s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Nmap done: 1 IP address (1 host up) scanned in 0.13 seconds
```

Figure 1. Nmap host discovery of scanme.nmap.org.

4. Service and Attack-Surface Discovery

The service/version scan identified several network services exposed by the target. Nmap reported 22/tcp SSH, 80/tcp HTTP, 9929/tcp nping-echo, and 31337/tcp tcpwrapped as open. Ports 25/tcp, 139/tcp, and 445/tcp were reported as filtered. Nmap identified the operating system family as Linux.

Port	State	Service	Detected Version / Detail
22/tcp	Open	SSH	OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13
25/tcp	Filtered	SMTP	No version identified
80/tcp	Open	HTTP	Apache httpd 2.4.7 (Ubuntu)
139/tcp	Filtered	NetBIOS-SSN	No version identified
445/tcp	Filtered	Microsoft-DS	No version identified
9929/tcp	Open	nping-echo	Nping echo
31337/tcp	Open	tcpwrapped	Service protected/wrapped; no product version reported

```
(root@kali)-[/home/kali]
# nmap -sV scanme.nmap.org
Starting Nmap 7.94 ( https://nmap.org ) at 2026-09-09 12:56 PDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.083s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 993 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
25/tcp    filtered smtp
80/tcp    open  http     Apache httpd 2.4.7 ((Ubuntu))
139/tcp    filtered netbios-ssn
445/tcp    filtered microsoft-ds
9929/tcp  open  nping-echo Nping echo
31337/tcp open  tcpwrapped
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 9.54 seconds
```

Figure 2. Nmap service and version enumeration results.

5. Vulnerability Assessment Results

The terminal vulnerability scan used Nmap's vulnerability NSE scripts together with service detection. The output associated OpenSSH 6.6.1p1 and Apache httpd 2.4.7 with vulnerability references through the Vulners NSE output. The results below should be treated as potential

findings requiring validation because version/CPE matching does not by itself prove that every listed CVE is exploitable on the specific host.

ID	Observed Finding	Evidence from Terminal	Reported Score	Assessment Interpretation
V-01	Potential OpenSSH vulnerability match	OpenSSH 6.6.1p1; CVE-2023-38408 listed by Vulners	9.8	High-severity potential finding based on the detected OpenSSH version; validate applicability before calling it confirmed.
V-02	Potential OpenSSH vulnerability match	OpenSSH 6.6.1p1; CVE-2016-1908 listed by Vulners	9.8	High-severity potential finding based on version/CPE association; validate against the actual package and configuration.
V-03	Potential Apache vulnerability match	Apache httpd 2.4.7; Vulners output included an entry scored 10.0	10.0	Critical-severity potential finding requiring validation; the terminal output associates it with the detected Apache version.
V-04	Potential Apache vulnerability match	Apache httpd 2.4.7; Vulners output included entries scored 9.8	9.8	High-severity potential finding requiring validation.
V-05	Possible CSRF	Nmap http-csrf identified two search forms on / with /search/ actions	Not assigned by Nmap	Possible application-level issue; requires manual validation before being treated as a confirmed vulnerability.
V-06	Potential directory listing / interesting directory	Nmap http-enum identified /images/ as potentially interesting with listing	Not assigned by Nmap	Possible information-disclosure or unnecessary directory-exposure issue; validate the directory's contents and configuration.

6. Specific Terminal Evidence

The vulnerability-script output showed the following important evidence. Nmap identified the OpenSSH CPE as `cpe:/a:openbsd:openssh:6.6.1p1` and listed multiple vulnerability references, including CVE-2023-38408 and CVE-2016-1908, each with a reported 9.8 score in the terminal output. For Apache, Nmap identified `cpe:/a:apache:http_server:2.4.7` and displayed vulnerability references including an entry scored 10.0 and other entries scored 9.8.

The same vulnerability scan reported a possible CSRF condition for two forms on the site's root page: the `nst-head-search` form and the `nst-foot-search` form, both using `/search/` as the form action. Nmap also reported that no DOM-based XSS was found by the corresponding script. Finally, `http-enum` identified `/images/` as a potentially interesting directory with listing.

7. Supporting Scan Evidence

```
(root@kali)-[/home/kali]
# nmap -sV scanme.nmap.org -oN scanme-services.txt
Starting Nmap 7.94 ( https://nmap.org ) at 2026-09-09 12:58 PDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.078s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 993 closed tcp ports (reset)
PORT      STATE      SERVICE      VERSION
22/tcp    open      ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
25/tcp    filtered  smtp
80/tcp    open      http         Apache httpd 2.4.7 ((Ubuntu))
139/tcp    filtered  netbios-ssn
445/tcp    filtered  microsoft-ds
9929/tcp   open      nping-echo   Nping echo
31337/tcp  open      tcpwrapped
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.50 seconds
```

Figure 3. Saved Nmap service scan output using `-oN scanme-services.txt`.

8. Vulnerability Classification

The assessment distinguishes between confirmed observations and potential vulnerabilities. Open ports and software versions are exposure findings. The Vulners results are version/CPE-based vulnerability references, so they should be validated before being reported as confirmed CVEs. The possible CSRF and directory-listing observations are also preliminary because the Nmap scripts identify conditions that warrant further investigation.

The strongest documented risks are the potential OpenSSH and Apache vulnerability matches because the terminal output provides explicit vulnerability references and severity scores.

9. Security Implications

Exposed SSH provides a remotely reachable administrative service and should be kept patched and appropriately restricted.

An outdated web server version can increase the likelihood of exposure to known security issues if the associated vulnerability references apply.

Possible CSRF weaknesses can affect the integrity of actions performed through web forms if they are confirmed.

A directory listing may disclose files or information that was not intended to be publicly browsable.

Filtered ports indicate that some services may be protected by network filtering, reducing direct exposure compared with an open state.

10. Conclusion

The Nmap TestMe assessment demonstrated the major vulnerability-assessment techniques required for the project: asset discovery, service and version enumeration, vulnerability-oriented NSE scanning, and preservation of scan results. The assessment identified an active host at 45.33.32.156 and exposed services including SSH and HTTP. The vulnerability scan produced multiple potential vulnerability references associated with OpenSSH 6.6.1p1 and Apache httpd 2.4.7, as well as possible CSRF and directory-exposure observations. These results provide documented evidence for vulnerability assessment while maintaining the distinction between scanner-detected potential findings and vulnerabilities that have been independently confirmed.